

Evidence on a Silver Platter: Finding versus Using Evidence in Agentic Search

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Abstract

When an agentic search system gives a wrong answer, the failure is usually treated as a single outcome. Yet two very different problems may lie behind it: the agent may never find the necessary evidence, or it may find that evidence and still fail to use it. Existing end-to-end evaluations of browsing agents [Nakano et al., 2021, Yao et al., 2023] largely entangle these possibilities. I separate them through a controlled intervention on 100 BrowseComp-Plus questions [Chen et al., 2025], injecting annotated evidence into an otherwise fixed ReAct-style search agent. Providing annotated evidence closes most of the performance gap, while a single-shot reader with longer evidence views offers little additional benefit over live agentic access. Evidence discovery is the dominant bottleneck for this system. Disabling search after injection reduces accuracy, delaying injection changes how the agent spends its actions, and inserting evidence into a frozen trajectory is less effective than letting the agent respond to it dynamically.

1 Introduction

Agentic search systems built around large language models answer open-domain questions by interleaving reasoning steps with search and document-reading actions [Nakano et al., 2021, Yao et al., 2023]. On hard benchmarks such as BrowseComp [Wei et al., 2025], these systems frequently fail, but a raw accuracy number does not say *why*: does the agent fail to locate the documents that contain the answer, or does it locate them and still fail to extract or reason over the answer correctly? These two failure modes call for very different fixes - better retrieval and search policies versus better reading and reasoning - as retrieval-augmented generation research likewise separates external knowledge access from generation over retrieved context [Lewis et al., 2020]. Separating the two is therefore the central motivation for this project.

Distinguishing these explanations from ordinary agent logs is difficult. A failed trajectory may never retrieve a relevant document, may retrieve it only as a short preview, may open it but overlook the decisive passage, or may extract the

relevant facts but combine them incorrectly. Conversely, a correct answer does not reveal whether the search policy was effective or merely fortunate. Post-hoc inspection therefore mixes together retrieval quality, document presentation, context position, tool-use decisions, and final-answer reasoning. prior work shows that even the position of relevant evidence within a long context can materially affect model performance [Liu et al., 2023]. A controlled intervention is needed to determine how performance changes when the evidence-discovery problem is removed while the rest of the system remains as stable as possible.

I address this problem with an evidence-injection design built on BrowseComp-Plus, whose fixed corpus and evidence annotations support reproducible interventions [Chen et al., 2025]. I evaluate a fixed agent on the same 100 questions while systematically varying *whether*, *when*, and *how* annotated evidence documents become available. The live conditions compare an unmodified search agent with beginning and mid-trajectory injection, a single-shot full-evidence reader, and an injected agent without further search access. A complementary counterfactual replay places evidence into frozen no-injection trajectories, allowing me to test evidence use without permitting the search policy to adapt to the new observations. Across conditions, the model, retriever, corpus, question set, and grading procedure remain fixed. Before analysis, I verified the 100-question manifest, the alignment of question IDs across conditions, and the saved trajectories and intervention records to confirm that each run followed its intended configuration.

I use controlled evidence injection to measure how discovery, injection timing, continued search access, and trajectory adaptation affect the same agent on the same questions.

1.1 Research Questions

1. How much of the agent’s failure is caused by evidence discovery (i.e. never finding the right documents), as opposed to failing after the evidence was already available?
2. Does the *timing* of evidence injection within the trajectory affect accuracy or efficiency?
3. How do continued search access and the manner of evidence presentation (live agentic access vs. a frozen, distributed replay) affect how well the agent utilizes evidence it has been given?

1.2 Related Work

BrowseComp [Wei et al., 2025] established that hard-to-find, easy-to-verify questions are a severe challenge for browsing agents, with unaided models scoring near zero. BrowseComp-Plus [Chen et al., 2025] converts this into a fair, reproducible testbed by fixing a static, human-verified document corpus and retriever, so that retrieval and reasoning contributions can be isolated - the property this project relies on to construct controlled evidence-injection and evidence-removal conditions. This search agent follows the ReAct pattern

of interleaved reasoning and tool actions [Yao et al., 2023]. The finding that evidence *position* within a model’s context materially affects whether it is used, independent of whether the evidence is present at all, echoes Liu et al. [2023]’s observation that language models use different parts of a long context unevenly. my injection-timing (Section 3.2) and distributed-replay (Section 3.4) results can be read as an agentic-trajectory analogue of that effect.

Recent work also distinguishes retaining information from using it during later actions. MemoryArena evaluates memory inside interdependent agent–environment loops rather than as isolated recall [He et al., 2026]. MERMAID couples iterative retrieval with a persistent evidence memory for verification tasks [Cao et al., 2026]. TRACE evaluates efficiency and evidence grounding over complete research trajectories rather than relying only on final accuracy [Chen et al., 2026]. this motivates the behavioral and citation measures used here. Training work such as DR Tulu’s evolving, search-grounded rubrics studies on-policy supervision [Shao et al., 2025], whereas my experiments hold the policy fixed and intervene on inference-time evidence access.

2 Experimental Setup

2.1 Benchmark and agent

All experiments use BrowseComp-Plus’s validation data [Chen et al., 2025]. The search agent is a ReAct-style loop over a JSON action schema (`search` / `get_document` / `final`). The agent language model is GPT-4o (temperature 0) [Hurst et al., 2024]. Documents are retrieved with a Qwen3 dense retriever (Qwen3-Embedding-4B, top- $k = 5$, 512-token previews, normalized embeddings, EOS pooling) [Zhang et al., 2025] against BrowseComp-Plus’s static corpus [Chen et al., 2025]. Answers are scored with an LLM judge that reuses BrowseComp-Plus’s own grading template rather than a custom rubric [Chen et al., 2025].

2.2 Evidence documents and injection format

For each question, BrowseComp-Plus provides human-verified supporting-document annotations [Chen et al., 2025]. I fall back to gold-labeled documents when evidence annotations are unavailable. The confirmatory questions contain between 2 and 11 injected documents (mean 6.19. median 6). Conditions B, C, and E (table 1) show at most 512 tokens from each document, matching an ordinary search-result preview. The documents are deterministically shuffled using the experiment seed and question ID, assigned descending synthetic scores, and rendered as one ordinary search observation whose query is the original question. The synthetic turn appears before the first agent action in B and E, or after two real actions in C, and does not consume the action budget. The agent may subsequently open any visible document through `get_document`, which returns at most 4,096 tokens.

The nominal preview payload per injected turn is therefore capped between

1,024 and 5,632 tokens, with a median cap of 3,072 tokens. Exact realized token totals were not logged because documents shorter than the cap are not padded. Condition D uses the same document IDs and deterministic order but supplies up to 4,096 tokens per document in a dedicated single-shot prompt, for a nominal cap between 8,192 and 45,056 tokens. Consequently, D changes evidence length and interaction format together and cannot isolate either factor.

2.3 Conditions

Five conditions were run on the same 100 questions, holding the agent, retriever, and grading procedure fixed (Table 1):

Table 1: Live experimental conditions. Preview injection supplies at most 512 tokens per evidence document. the reader receives up to 4,096 tokens per document.

Condition	Evidence access	Available tools	Budget
A: no injection	None. evidence must be discovered	Search and open	10
B: beginning	Preview injection before action 1	Search and open	8
C: mid-trajectory	Preview injection after action 2	Search and open	8
D: full-evidence reader	Longer evidence views in a single prompt	None	0
E: search disabled	Preview injection before action 1	Open only	8

Figure 1 summarizes the intervention used in these five conditions. Evidence injection bypasses retrieval by inserting the annotated documents directly into the trajectory as a search-result observation. Conditions B and E receive this observation before the first agent decision, whereas C first takes two ordinary actions. After injection, the live agent can continue choosing actions within its remaining budget. E removes only the `search` action, so the agent may still open an injected document with `get_document`. Condition D instead receives the evidence in a separate single-shot prompt and answers without tools. The diagram is schematic: action boxes can represent multiple search or document-reading steps, and the offline R0/R-all replay is described separately below.

2.4 Distributed-evidence counterfactual replay

As a sixth, offline condition, I freeze condition A’s own no-injection trajectories. **R0** reuses the original condition-A answers and metrics. I do not regenerate answers for an unchanged trajectory. **R-all** distributes every annotated evidence document into the trajectory’s existing search-result slots. The placement procedure shuffles evidence IDs, search turns, and eligible ranks with a fixed seed, then assigns one missing evidence document per turn per pass. It never replaces a document that the original agent opened or cited, preserves the displaced result’s score, and uses the same 512-token preview format as live search. The final answer is regenerated once from a prompt containing the question and complete frozen session record, with tools disabled.

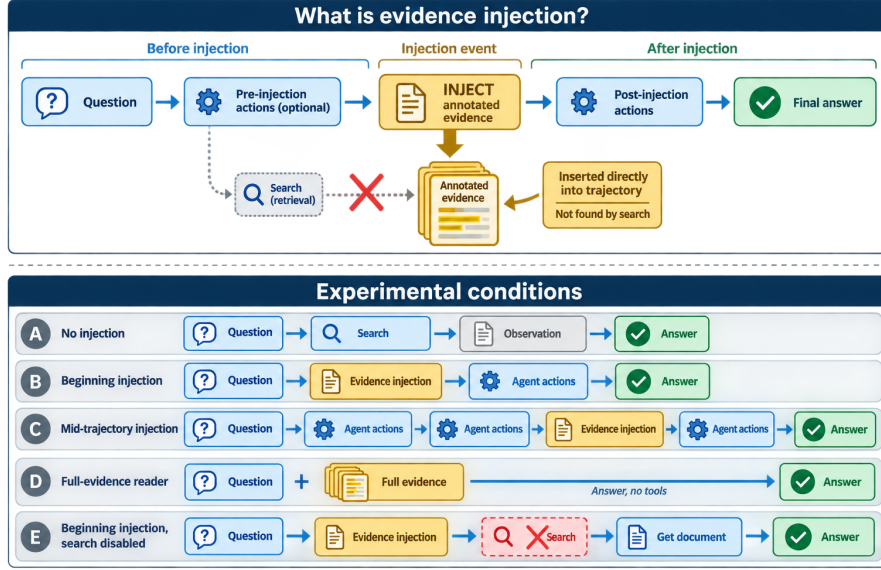


Figure 1: Overview of evidence injection and live experimental conditions A–E. The intervention places annotated evidence directly into the agent’s trajectory rather than requiring the search policy to retrieve it. The timing of this insertion and the tools available afterward vary by condition.

Because the action sequence is frozen at its original, off-policy state, later search actions were generated under the original observations and may not be behaviorally consistent with the injected evidence. This removes evidence-conditioned search-policy adaptation from R-all but introduces differences in trajectory rendering and final-answer prompting relative to R0 and the live conditions. Seed 43 is the complete primary replay. Two interrupted exploratory runs produced partial results for seeds 42 and 44. these are analyzed only on the 47 questions completed under all three seeds.

2.5 Metrics and statistical tests

The primary metric is answer accuracy under the LLM judge [Zheng et al., 2023], with 95% bootstrap confidence intervals (question-level resampling, 10,000 resamples) reported for each condition in isolation. Because every condition shares the same 100 questions, cross-condition comparisons use paired statistics instead: an exact McNemar test on the paired correctness contingency table [McNemar, 1947], and paired bootstrap confidence intervals [Efron, 1979] (question IDs resampled together, 10,000 resamples) for efficiency metrics such as actions-to-answer and search-call counts. This paired analysis makes each contrast reflect within-question changes rather than differences in question difficulty.

As a direct behavioral proxy for evidence use, I also record whether the structured final action cites at least one injected document (or, for R-all, an

annotated evidence document present in the replay). This measure establishes explicit reference to supplied evidence, but it does not test whether every answer claim is entailed by the cited text.

3 Results

3.1 Evidence discovery is the dominant bottleneck

Beginning injection raised accuracy from 18% to 72% (+54 points; McNemar $p = 1.2 \times 10^{-14}$), with 56 questions changing from incorrect to correct and two changing in the opposite direction (Figure 2). It also reduced mean actions from 8.05 to 3.05 and search calls from 5.45 to 1.06. Most failures in this configuration therefore arose from failing to find the evidence, rather than using it once available (Table 2).

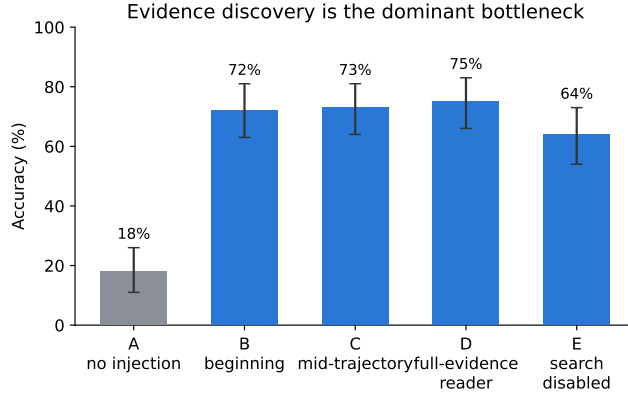


Figure 2: Accuracy across the five live conditions ($n = 100$ each; 95% bootstrap CIs). Providing evidence raises accuracy by 54 points, while the three conditions with live or full evidence access cluster between 72% and 75%. Disabling further search reduces this to 64%.

3.2 Injection timing redistributes effort but does not affect accuracy

Beginning and mid-trajectory injection produced similar accuracy (72% vs. 73%; McNemar $p = 1.0$; discordant pairs 3 vs. 4; Figure 3). Compared with C, B used 1.13 fewer searches (95% CI $[-1.41, -0.83]$) and 1.08 fewer actions (CI $[-1.39, -0.75]$) overall, but 0.71 more searches (CI $[0.45, 1.00]$) and 0.92 more actions (CI $[0.61, 1.25]$) after injection. Early evidence therefore reduced total effort while shifting more work to post-injection verification.

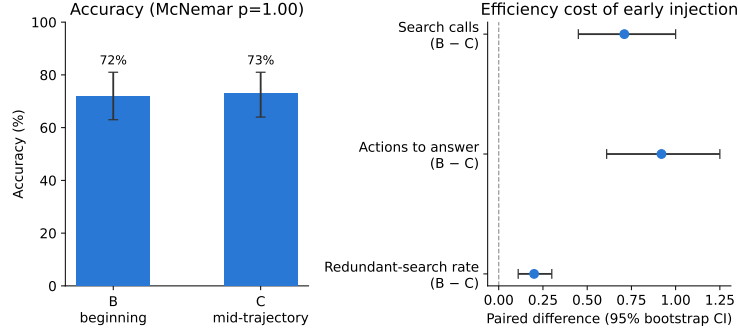


Figure 3: Injection timing changes behavior, not accuracy. Left: B and C differ by only one accuracy point. Right: negative values show that beginning injection uses fewer actions and searches overall. positive values show that it performs more verification after evidence becomes visible.

3.3 Continued search still contributes once evidence is present

Retaining search after injection added 8 accuracy points ($E \rightarrow B$; $p = 0.02$), although the experiment cannot distinguish new evidence from verification or disambiguation. The full-evidence reader added only 3 points over B and the difference was not significant ($p = 0.51$; discordant pairs 3 vs. 6). Thus, the dominant effect was evidence injection itself (+54 points), while live and single-shot evidence access performed similarly (Figure 4).

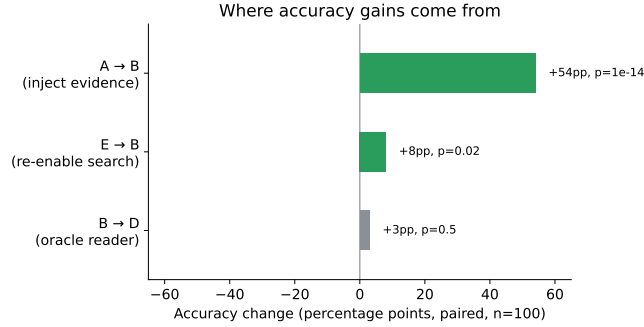


Figure 4: Paired accuracy gains on the same 100 questions. Evidence injection is the dominant intervention (+54 points); retaining search contributes a smaller but significant +8 points; the +3-point reader difference is not significant.

3.4 Distributed evidence utilization under a frozen, off-policy trajectory

Distributed evidence raised frozen-replay accuracy from 18% to 53% (+35 points; 95% CI [25, 45]; McNemar $p = 2.84 \times 10^{-9}$), rescuing 37 questions and harming

two (Figure 5). R-all nevertheless remained 19, 20, and 22 points below B, C, and D, respectively (all $p < 0.002$). Evidence therefore helped even in a frozen trajectory, but the remaining gap cannot distinguish off-policy presentation, absent policy adaptation, and replay-prompt differences.

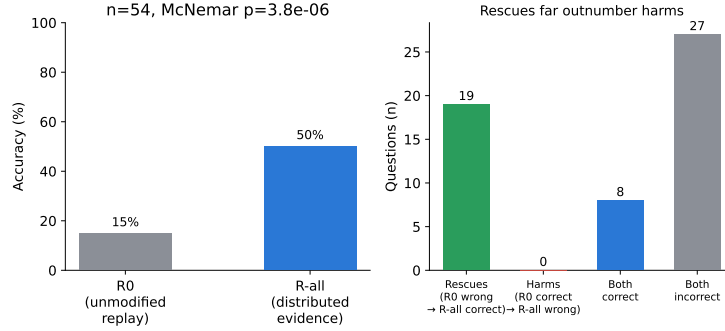


Figure 5: Distributed evidence improves the frozen-trajectory evaluation on the full 100-question set. Left: accuracy rises from 18% to 53%. Right: paired outcomes show 37 rescues and only 2 harms.

3.5 Master results table

Table 2: Accuracy by condition ($n = 100$ each). R0 reuses A.

Condition	Accuracy	95% CI
A: no injection	18.0%	[11.0, 26.0]
B: beginning	72.0%	[63.0, 81.0]
C: mid-trajectory	73.0%	[64.0, 81.0]
D: full-evidence reader	75.0%	[66.0, 83.0]
E: search disabled	64.0%	[54.0, 73.0]
R0: original A	18.0%	[11.0, 26.0]
R-all: distributed replay	53.0%	[43.0, 63.0]*

* Question-level bootstrap CI.

4 Discussion and Limitations

Providing evidence resolved 56 previously incorrect questions and harmed two, identifying evidence discovery as the main bottleneck in this configuration. The full-evidence reader gained only three points over live access, while disabling search cost eight points and frozen replay remained 19–22 points below the live conditions. Improving retrieval is therefore the clearest next step, although evidence delivery and policy adaptation still matter.

4.1 Broader implications: evidence shadowing

The replay result shows that placing evidence in a trajectory is not equivalent to letting the agent respond to it. I use *evidence shadowing* only as shorthand for this gap: the experiment does not separate document position, presentation, additional context, or absent evidence-conditioned actions.

Limitations and future work. Results cover one model, retriever, and 100-question sample. Conditions A and B use 10- and 8-action budgets, respectively; this cannot explain B’s accuracy gain but prevents a matched efficiency comparison. Conditions B and D also differ in evidence length and interaction format. The LLM judge received documented manual corrections only for three condition-A cases and was not checked against an alternative grader. Priority follow-ups are a budget-matched control, controlled evidence-position experiments, additional placement seeds, and replication with another model and grader.

5 Code

GitHub Repo: <https://github.com/AlonBarak-dev/evidence-on-silver-platter>

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